

ROBERTO TEJADA AREVALO
CURRICULUM VITAE

Department of Astrophysical Sciences
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Education:

- Ph.D. in Astrophysical Sciences, Princeton University (2026).
- B.S. Physics, California State University, Los Angeles (2020), Summa Cum Laude.
- A.A. Physics and Mathematics, Los Angeles City College (2017).

Selected Publications:

- **Tejada Arevalo, R.**, Burrows, A., Sur, A., Su, Y. *ORCHARD: A General Planetary Evolution Code*. Submitted to ApJ. arXiv:2604.24845
- **Tejada Arevalo, R.**, Gupta, A., Burrows, A., Zheng, D., Tang, Y., Deng, J. *Sub-Neptune Memories I: Implications of Inefficient Mantle Cooling and Silicate Rain*. 2026. ApJ, 1001, 243.
- **Tejada Arevalo, R.** *Different Inhomogeneous Evolutionary Histories for Uranus and Neptune*. 2025, ApJL, 989, L40.
- **Tejada Arevalo, R.**, Sur, A., Su, Y., Burrows, A. *Jupiter Evolutionary Models Incorporating Stably Stratified Regions*. 2025, ApJ, 979, 243
- Sur, A. **Tejada Arevalo, R.**, Burrows, A., Chen, Y. *Next-Generation Improvements in Giant Exoplanet Evolutionary and Structural Models*. Accepted to ApJ. arXiv:2510.08681
- Sur, A., **Tejada Arevalo, R.**, Burrows, A., Su, Y. *The Evolution of Jupiter and Saturn as a function of the Semi-convective parameter R_p* . Accepted to ApJL. arXiv:2506.19041.
- Sur, A., **Tejada Arevalo, R.**, Su, Y., Burrows, A. *Simultaneous Evolutionary Fits for Jupiter and Saturn Incorporating Fuzzy Cores*. 2025, ApJL, 980, L5
- **Tejada Arevalo, R.**, Sur, A., Su, Y., Burrows, A. *Equations of State, Thermodynamics, and Miscibility Curves for Jovian Planet and Giant Exoplanet Evolutionary Models*. 2024 ApJS, 274, 34
- Sur, A., **Tejada Arevalo, R.**, Su, Y., Burrows, A., *APPLE: An Evolution Code for Modeling Giant Planets*. 2024, 971, 104.
- Chen, Y., Burrows, A., Sur, A., **Tejada Arevalo, R.** *Jupiter Atmospheric Models and Outer Boundary Conditions for Giant Planet Evolutionary Calculations*. 2023, ApJ, 957, 36
- **Tejada Arevalo, R.**, Tamayo, D., Cranmer, M. *Do Young Planets Form in Mean Motion Resonance Chains?* 2022, ApJL, 932, L2
- **Tejada Arevalo, R.**, Winn, J., Anderson, K. *Further Evidence of Tidal Spin-up For Hot Jupiter Host β Stars*. 2021, ApJ, 919, 2.

Awards:

- May 2024: Princeton Astrophysical Sciences Dept. Mentorship/Diversity Award

- March 2022: Ford Foundation Predoctoral Fellowship

References:

- **Adam S. Burrows**
Professor of Astrophysical Sciences, Princeton University
burrows@astro.princeton.edu
- **Josh N. Winn**
Professor of Astrophysical Sciences, Princeton University
jwinn@princeton.edu
- **Jonathan Fortney**
Chair and Professor of Astrophysics, University of California, Santa Cruz
jfortney@ucsc.edu

Recent Talks:

- March 16, 2026: AASTCS11 Conference (Talk)
- December 19, 2025: AGU 2025 Conference (Talk)
- November 25, 2025: Caltech GPS Seminar
- November 10, 2025: UC Santa Cruz Planetary Lunch Seminar
- November 5, 2025: Yale Exoplanet and Stars Seminar
- October 21, 2025: MIT EAPS Planetary Lunch Seminar
- October 17, 2025: Caltech TAPIR Seminar
- October 16, 2025: UC San Diego Stars and Planets Seminar
- October 13, 2025: Caltech Tea Talk
- September 2025: UCLA Tuesday Lunch Talk
- September 2025: Princeton Astrophysical Sciences Bahcall Lunch (Talk)
- July 2025: Other Worlds Laboratory Conference, UC Santa Cruz (Talk)
- June 2025: AAS 246 (Dissertation Talk)
- May 2025: NY Area Exoplanet Meetup (Talk)

Calbridge

I was a Calbridge Undergraduate Scholar and now serve in the Calbridge Alumni Committee. I participate in the Calbridge PhD program by volunteering to lead coding and presentation undergraduate workshops.

Astrophysical Sciences Peer-Mentorship Program

I led our department's internal peer-mentorship program for two years. I worked with graduate students and Postdocs to pair them in mentorship pairs according to the graduate students' needs. Junior graduate students (years 1 & 2) were paired with Senior graduate students (years 3 and above), and Senior graduate students were paired with Postdocs.